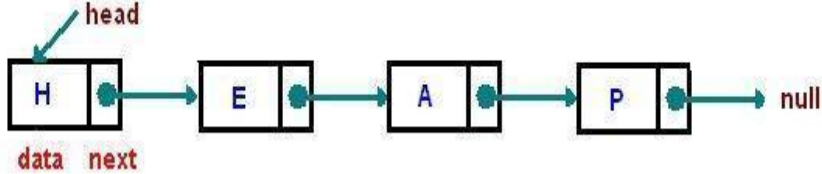
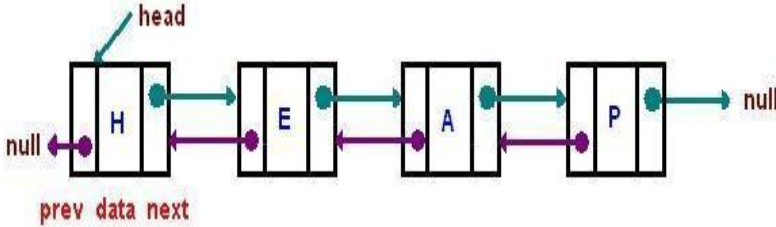
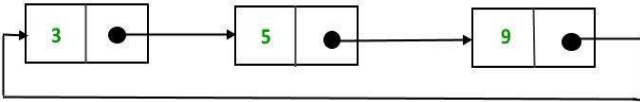


**Unit 1**  
**ABSTRACT DATATYPE**

**PART A**

| Q.N | Question                                                                                                                                                                                                                                                                                           | Bloom's Taxonomy Level |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1   | <b>Define Data Structures</b><br>Data Structures is defined as the way of Organizing all data items that consider not only the elements stored but also stores the relationship between the elements.                                                                                              | K1                     |
| 2   | <b>Define Linked Lists</b><br>Linked list consists of a series of structures, which are not necessarily adjacent in memory. Each structure contains the element and a pointer to a structure containing its successor. We call this the Next Pointer. The last cell's Next pointer points to NULL. | K1                     |
| 3   | <b>State the different types of linked lists</b><br>The different types of linked list include singly linked list, doubly linked list circular linked list.                                                    | K1                     |
| 4   | <b>How the singly linked lists can be represented?</b>  <p>Each node has two elements</p> <ul style="list-style-type: none"> <li>i) Data</li> <li>ii) Next</li> </ul>                                          | K1                     |
| 5   | <b>How the doubly linked list can be represented?</b>                                                                                                                                                          | K1                     |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 6  | <p><b>When singly linked list can be represented as circular linked list?</b></p> <p>In a singly linked list, all the nodes are connected with forward links to the next nodes in the list. The last node has a next field, NULL. In order to implement the circularly linked lists from singly linked lists, the last node's next field is connected to the first node.</p>                                 | K1 |
| 7  | <p><b>List down any four applications of data structures?</b></p> <ul style="list-style-type: none"> <li>• Compiler design</li> <li>• Operating System</li> <li>• Database Management system</li> <li>• Network analysis</li> </ul>                                                                                                                                                                                                                                                           | K1 |
| 8  | <p><b>What is a circular linked list?</b></p> <p>A circular linked list is a special type of linked list that supports traversing from the end of the list to the beginning by making the last node point back to the head of the list.</p>                                                                                                                                                                                                                                                   | K1 |
| 9  | <p><b>List the basic operations carried out in a linked list</b></p> <p>The basic operations carried out in a linked list include:</p> <ul style="list-style-type: none"> <li>• Creation of a list</li> <li>• Insertion of a node</li> <li>• Deletion of a node</li> <li>• Modification of a node</li> <li>• Traversal of the list</li> </ul>                                                                                                                                                 | K1 |
| 10 | <p><b>List out the advantages of using a linked list</b></p> <ul style="list-style-type: none"> <li>• It is not necessary to specify the number of elements in a linked list during its declaration</li> <li>• Linked list can grow and shrink in size depending upon the insertion and deletion that occurs in the list</li> <li>• Insertions and deletions at any place in a list can be handled easily and efficiently</li> <li>• A linked list does not waste any memory space</li> </ul> | K1 |
| 11 | <p><b>List out the disadvantages of using a linked list</b></p> <ul style="list-style-type: none"> <li>• Searching a particular element in a list is difficult and time consuming</li> <li>• A linked list will use more storage space than an array to store the same number of elements</li> </ul>                                                                                                                                                                                          | K1 |
| 12 | <p><b>Mention three examples that uses linked list?</b></p> <ul style="list-style-type: none"> <li>• Polynomial ADT</li> <li>• Radix sort</li> </ul>                                                                                                                                                                                                                                                                                                                                          | K1 |

|                                                                             | <ul style="list-style-type: none"><li>Multi lists</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------|----------------------------|----------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------|-------------------------|----|
| 13                                                                          | <p><b>State the difference between arrays and linked lists</b></p> <table><thead><tr><th>Arrays</th><th>Linked Lists</th></tr></thead><tbody><tr><td>Size of an array is fixed.</td><td>Size of a list is variable</td></tr><tr><td>It is necessary to specify the number of elements during declaration.</td><td>It is not necessary to specify the number of elements during declaration.</td></tr><tr><td>Insertions and deletions are somewhat difficult.</td><td>Insertions and deletions are carried out easily.</td></tr><tr><td>It occupies less memory than a linked list for the same number of elements.</td><td>It occupies more space.</td></tr></tbody></table> | Arrays | Linked Lists | Size of an array is fixed. | Size of a list is variable | It is necessary to specify the number of elements during declaration. | It is not necessary to specify the number of elements during declaration. | Insertions and deletions are somewhat difficult. | Insertions and deletions are carried out easily. | It occupies less memory than a linked list for the same number of elements. | It occupies more space. | K4 |
| Arrays                                                                      | Linked Lists                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| Size of an array is fixed.                                                  | Size of a list is variable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| It is necessary to specify the number of elements during declaration.       | It is not necessary to specify the number of elements during declaration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| Insertions and deletions are somewhat difficult.                            | Insertions and deletions are carried out easily.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| It occupies less memory than a linked list for the same number of elements. | It occupies more space.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| 14                                                                          | <p><b>Define an Abstract Data Type (ADT)</b><br/>An abstract data type is a set of operations. ADTs are mathematical abstractions; nowhere in an ADT's definition is there any mention of how the set of operations is implemented. Objects such as lists, sets and graphs, along with their operations can be viewed as abstract data types.</p>                                                                                                                                                                                                                                                                                                                             | K1     |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| 15                                                                          | <p><b>What is single linked list?</b><br/>It is a linear data structure which consists a pointer field that points to the address of its next node (successor) and the data item.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | K1     |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| 16                                                                          | <p><b>Define double linked list?</b><br/>It is linear data structure which consists of two links or pointer fields</p> <ul style="list-style-type: none"><li>• <i>Next</i> pointer points to the address of the next(successor) node.</li><li>• <i>Previous</i> pointer points to the address of the previous(predecessor) node.</li></ul>                                                                                                                                                                                                                                                                                                                                    | K1     |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| 17                                                                          | <p><b>What are the two basic types of Data structures?</b></p> <ol style="list-style-type: none"><li>1. Primitive Data structure<br/>Eg., int,char,float</li><li>2. Non Primitive Data Structure<ol style="list-style-type: none"><li>i. Linear Data structure<br/>Eg., Lists Stacks Queues</li><li>ii. Non linear Data structure<br/>Eg., Trees Graphs</li></ol></li></ol>                                                                                                                                                                                                                                                                                                   | K1     |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |
| 18                                                                          | <p><b>Define circularly linked list with neat diagram.</b><br/>Circularly linked list is a collection of nodes, where each node is a structure containing the element and a pointer to a structure containing its successor. The pointer field of the last node points to the address of the first node. Thus the linked list becomes circular.</p>                                                                                                                                                                                                                                                                                                                           | K1     |              |                            |                            |                                                                       |                                                                           |                                                  |                                                  |                                                                             |                         |    |

|           |                                                                                                                                                                                                                                                                                                                                                                         |           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>19</b> | <b>List the advantages of Array over Linked List.</b> <ol style="list-style-type: none"> <li>1. Array has a specific address for each element stored in it and thus we can access any memory directly.</li> <li>2. As we know the position of the middle element and other elements are easily accessible too, we can easily perform BINARY SEARCH in array.</li> </ol> | <b>K1</b> |
| <b>20</b> | <b>What is the need for the header?</b><br>Header of the linked list is the first element in the list and it stores the number of elements in the list. It points to the first data element of the list.                                                                                                                                                                | <b>K1</b> |

### PART B

| <b>Q.NO</b> | <b>Question</b>                                                                                                                                                                                                                                                                                                                                                        | <b>Bloom's Taxonomy Level</b> |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>1</b>    | <ol style="list-style-type: none"> <li>i. What is a singly linked list? (2)</li> <li>ii. Describe the suitable routine segments for any four operations. (14)</li> </ol>                                                                                                                                                                                               | <b>K2</b>                     |
| <b>2</b>    | List an algorithm to perform the following operations in a doubly linked list. <ol style="list-style-type: none"> <li>i. Insert a node at the end of the list.(8)</li> <li>ii. Delete the last node in the list. (8)</li> </ol>                                                                                                                                        | <b>K1</b>                     |
| <b>3</b>    | <ol style="list-style-type: none"> <li>i. Give the algorithm to perform insertion on given position on a doubly linked list.(8)</li> <li>ii. Give the algorithm to delete the given node on doubly linked list.(8)</li> </ol>                                                                                                                                          | <b>K1</b>                     |
| <b>4</b>    | Write an algorithm to demonstrate a polynomial using a linked list for Addition operation. (16)                                                                                                                                                                                                                                                                        | <b>K2</b>                     |
| <b>5</b>    | Analyze and write algorithm for Circular Linked list for the following operations using structure pointer. <ol style="list-style-type: none"> <li>(i) Create &amp; Insert . (8)</li> <li>(ii) Delete &amp; Display.(8)</li> </ol>                                                                                                                                      | <b>K4</b>                     |
| <b>6</b>    | Consider an array A[1: n] Given a position, write an algorithm to insert an element in the Array. If the position is empty, the element is inserted easily. If the position is already occupied the element should be inserted with the minimum number of shifts. (Note: The elements can shift to the left or to the right to make the minimum number of moves). (16) | <b>K1</b>                     |
| <b>7</b>    | Describe the various operations of the list ADT with examples. (16)                                                                                                                                                                                                                                                                                                    | <b>K2</b>                     |

|   |                                                                                                                                                                                                                                                             |    |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 8 | <p>Illustrate how polynomial manipulations are performed using lists? Explain addition operation with example. Apply the function for the following input.</p> <p><math>A=3x^{14}+2x^{13}+1</math> <math>B=8x^{14}+3x^{13}+3x^{12}+10x^{11}</math> (16)</p> | K2 |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|